

SESSION_04

Q.1 Create a simple JavaScript function called calculateTotal that takes two numbers: itemPrice and quantity, and returns the total bill amount using arithmetic operators.

A.1

```
#include <stdio.h>

float calculateTotal(float itemPrice, int quantity)
{
    return itemPrice * quantity;
}

int main() {
    float itemPrice = 500.0;
    int quantity = 3;

    float total = calculateTotal(itemPrice, quantity);
    printf("Total Bill: %.2f\n", total);
    return 0;
}
```

Q.2 Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean isMember, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).

A.2

```
#include <stdio.h>

int main() {

    float productPrice = 2000.0;
    float discountPercentage = 20.0;
    int isMember = 1;    // 1 = true, 0 = false

    float discount = productPrice * discountPercentage / 100;
    float finalPrice = productPrice - discount;

    if (isMember)
    {
```

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```
finalPrice = finalPrice - (finalPrice * 5 / 100);
}

printf("Product Price: %.2f\n", productPrice);
printf("Discount: %.2f%%\n", discountPercentage);
printf("Member: %d\n", isMember);
printf("Final Price: %.2f\n", finalPrice);

return 0;
}
```

Q.3 Write a function `isEligibleForOffer` that takes a user's age and total order value, and returns true if the user is 18 or older AND the order value is above 500, otherwise false. Use relational and logical operators together.

A.3

```
#include <stdio.h>

int isEligibleForOffer(int age, float orderValue)
{
    if (age >= 18 && orderValue > 500)
    {
        return 1;
    }
    else
    {
        return 0;
    }
}

int main() {
    int age = 20;
    float orderValue = 750.0
    int eligible = isEligibleForOffer(age, orderValue);

    printf("Eligible for offer: %d\n", eligible);
    return 0;
}
```

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Q.4 Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

```
A.4 #include <stdio.h>
    int main() {
        int likes = 1200;
        int comments = 150;
        int shares = 60;

        int trending = (likes >= 1000) ||
            (comments > 200 && shares >= 50);

        printf("Is the post trending? %d\n", trending);
        return 0;
    }
```

Q.5 Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.

```
A.5 #include <stdio.h>
    int main() {

        int followerCount = 100;

        printf("Before increment: %d\n", followerCount);
        printf("Post-increment: %d\n", followerCount++);
        printf("After post-increment: %d\n", followerCount);
        printf("Pre-increment: %d\n", ++followerCount);
        printf("After pre-increment: %d\n", followerCount);
        return 0;
    }
```

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